

Tool Calibration KPI Dashboard

End-to-end data project: raw records → Python cleaning → SQLite / SQL → interactive KPI dashboard

Sajin Saji · M.Eng Mechatronics & Cyber-Physical Systems, TH Deggendorf · October 2026 · github.com/sajin-saji/tool-calibration-kpi-dashboard

Note on the data: all records are randomly generated by `src/generate_data.py` for demonstration. They do not describe any real company. Errors were added on purpose so that the cleaning step has real work to do.

1. Goal

A tool center must make sure every measuring and assembly tool (torque wrenches, gauges, calipers and more) is calibrated on time. An overdue tool can lead to wrong measurements, rework and audit findings. This project answers four questions for the tool center team:

1. How many tools are past their calibration date, and in which department?
2. Which tools must be scheduled for calibration in the next 30 days?
3. Is calibration quality stable (first-time pass rate, turnaround time)?
4. Which tool types cause the highest calibration cost?

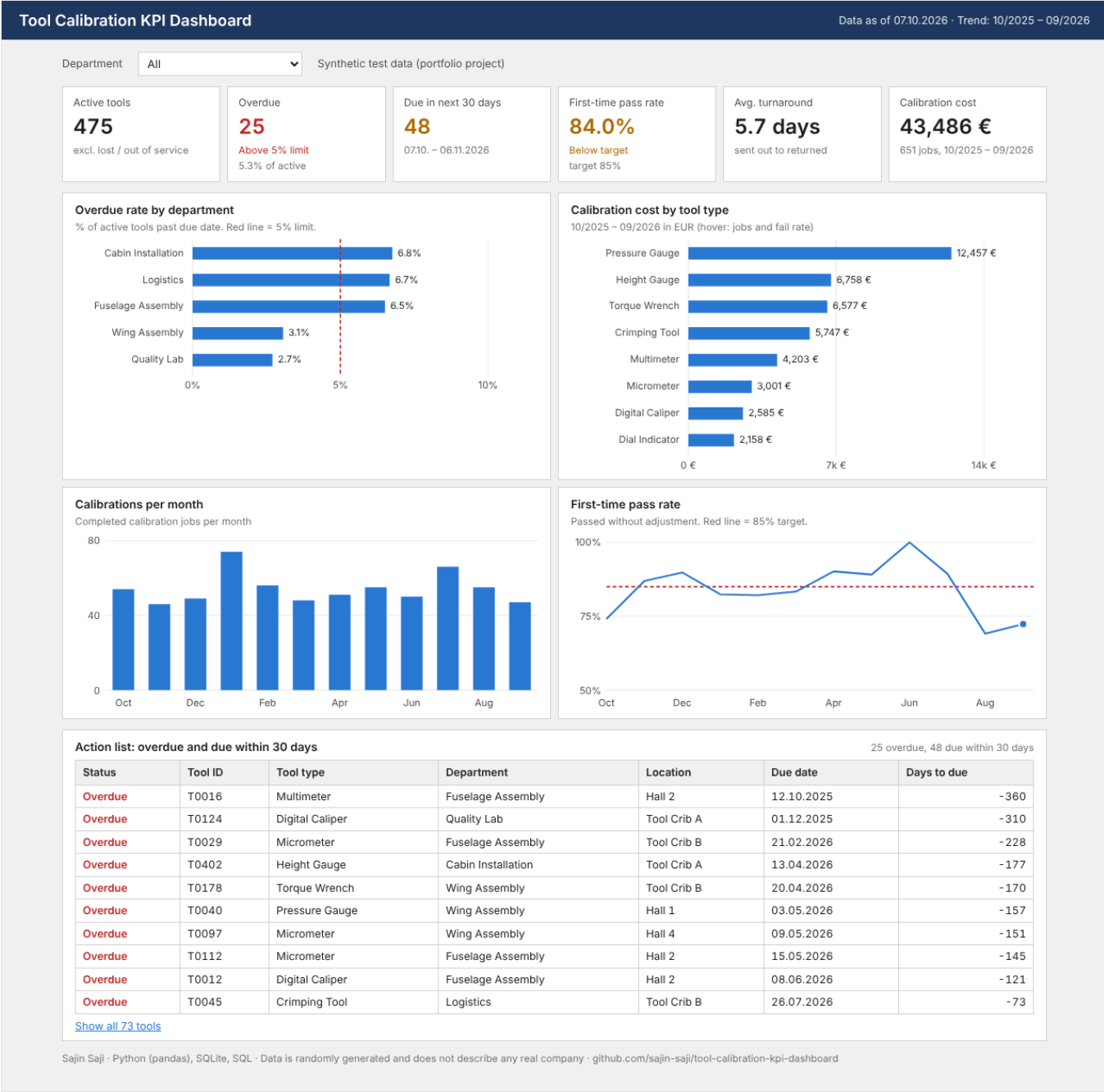
2. Results at a glance

Data as of 07.10.2026. Quality and cost figures cover 10/2025 – 09/2026.

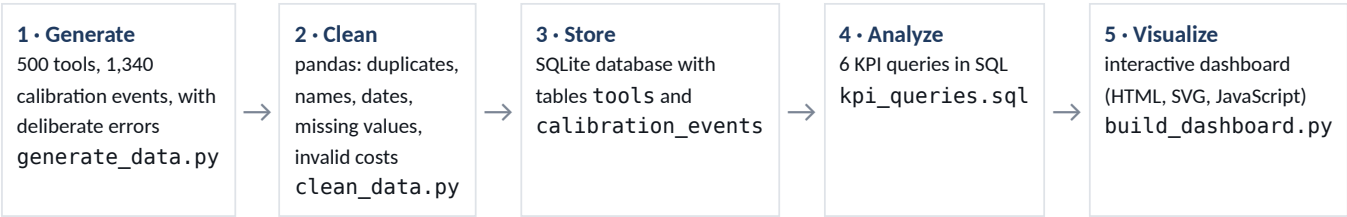
Active tools 475	Overdue (limit 5%) 25 · 5.3%	Due in next 30 days 48
First-time pass rate (target 85%) 84.0%	Average turnaround 5.7 days	Calibration cost (651 jobs) 43,486 €

3. Dashboard

Interactive version: [dashboard/index.html](#) (GitHub Pages). Department filter, hover tooltips and a full action list are available online.



4. Pipeline



5. Data cleaning

Problem found in raw data	Fix	Rows
Duplicate tool rows	Removed	8
Duplicate calibration events	Removed	15
Department typos (e.g. "QA Lab", "Fuselage assy")	Mapped to standard names	15
Tool types with mixed casing and spaces	Trimmed and title-cased	25
Dates in DD.MM.YYYY format	Converted to ISO (YYYY-MM-DD)	20
Missing next-due dates	Recomputed: last calibration + interval	12
Calibration events with zero or negative cost	Removed as invalid	10

Result: 500 clean tools and 1,315 clean calibration events. Every fix is logged in data/cleaning_report.txt.

6. KPI definitions

KPI	Definition
Active tools	All tools except status "Lost" or "Out of Service"
Overdue	Active tools with next-due date before the reporting date
Due in 30 days	Active tools due between 07.10. and 06.11.2026
First-time pass rate	Calibrations with result "Pass" (no adjustment, no failure) ÷ all calibrations
Turnaround	Average days from sending a tool to calibration until its return
Calibration cost	Sum of calibration costs in the trend window

Example: overdue rate by department (SQL)

```
SELECT department,
       COUNT(*) AS active_tools,
       SUM(next_due_date < '2026-10-07') AS overdue,
       ROUND(100.0 * SUM(next_due_date < '2026-10-07') / COUNT(*), 1) AS overdue_pct
FROM tools
WHERE status NOT IN ('Lost', 'Out of Service')
GROUP BY department
ORDER BY overdue_pct DESC;
```

7. Findings

Overdue rate by department				Cost by tool type (12 months)			
Department	Active	Overdue	Rate	Tool type	Jobs	Cost	Fail
Cabin Installation	74	5	6.8%	Pressure Gauge	135	12,457 €	4.4%
Logistics	60	4	6.7%	Height Gauge	63	6,758 €	1.6%
Fuselage Assembly	169	11	6.5%	Torque Wrench	101	6,577 €	5.0%
Wing Assembly	98	3	3.1%	Crimping Tool	109	5,747 €	5.5%
Quality Lab	74	2	2.7%	Multimeter	53	4,203 €	11.3%

- **25 of 475 active tools (5.3%) are overdue**, just above the 5% limit.
- The backlog is concentrated in **Cabin Installation, Logistics and Fuselage Assembly** (6.5–6.8%). Fuselage Assembly has the most overdue tools in absolute terms (11).
- **48 tools fall due within 30 days**. If they are not scheduled now, the overdue rate will rise further.
- The **first-time pass rate is 84.0%**, just below the 85% target, and fell to **69–72% in Aug-Sep 2026**.
- **Pressure gauges** are the largest cost driver (29% of cost); **multimeters** have the highest fail rate (11.3%).

8. Recommendations

1. **Clear the backlog first** in Fuselage Assembly (11 tools) and Cabin Installation, using the dashboard's action list sorted by days overdue.
2. **Book the 48 tools due within 30 days now**, so that they do not become overdue.
3. **Investigate the Aug-Sep drop in pass rate**: check whether specific tool types, a supplier or a lab caused it.
4. **Review multimeters** (highest fail rate): check usage and handling, or shorten their calibration interval.

9. Tools and repository

Tools: Python (pandas) · SQLite / SQL · HTML, SVG and JavaScript for the dashboard · Power BI rebuild guide in docs/power_bi_guide.md.

data/raw/	raw CSVs with deliberate errors
data/clean/	cleaned CSVs
data/calibration.db	SQLite database
sql/kpi_queries.sql	all KPI queries
src/	generate_data.py · clean_data.py · build_dashboard.py
dashboard/index.html	interactive dashboard (GitHub Pages)
docs/	project report, screenshot, Power BI guide